Automated Investment Data Reconciliation & Quality Pipeline

Python · SQL (SQLite) · pandas · openpyxl · Generative AI (Claude) with verification

Executive summary

Investment Operations teams reconcile their internal book of record (IBOR) against the custodian every day. Most differences are harmless timing effects, but they are mixed in with real problems such as unbooked trades, unprocessed corporate actions and bad prices, and sorting them by hand is slow and error-prone. This project automates that workflow end to end: it ingests messy source files, runs data-quality checks, reconciles positions in SQL, assigns a root cause to every break, writes a plain-English explanation (optionally with an LLM whose output is fact-checked), and delivers an Excel exception report and an HTML dashboard.

On one simulated month (21 business days, 3 client portfolios, 60 securities, 191 trades) the pipeline reconciled **2,407** position lines in about 1 second, matched **90.8%** automatically, explained **191** of 222 breaks as settlement timing, and narrowed the review queue to **31** exceptions (1.3% of lines), each with a root cause, severity and suggested action.

Measure	Result	How it is measured
Position lines reconciled	2,407	IBOR and custodian lines, full outer join
Auto-match rate	90.8%	Lines within quantity, price and value tolerances
Breaks auto-explained	191 of 222	Difference equals unsettled trades (cleared at settlement)
Exceptions for review	31	Breaks with a non-timing or secondary cause
Break detection (recall / precision)	100% / 100%	Against ground truth recorded when the data was generated
Root-cause accuracy	100% (222/222)	Assigned category equals injected category
Data-quality issues (recall / precision)	100% / 100%	10 injected issues
Robustness, 20 random seeds	4,544 breaks: 0 missed, 100% correct; DQ precision 99.0%	48,696 position lines in total

All data is synthetic: no real client, security or price data is used. The results show the logic works on the scenarios it was designed for; they are not a claim about production accuracy (see Limitations).

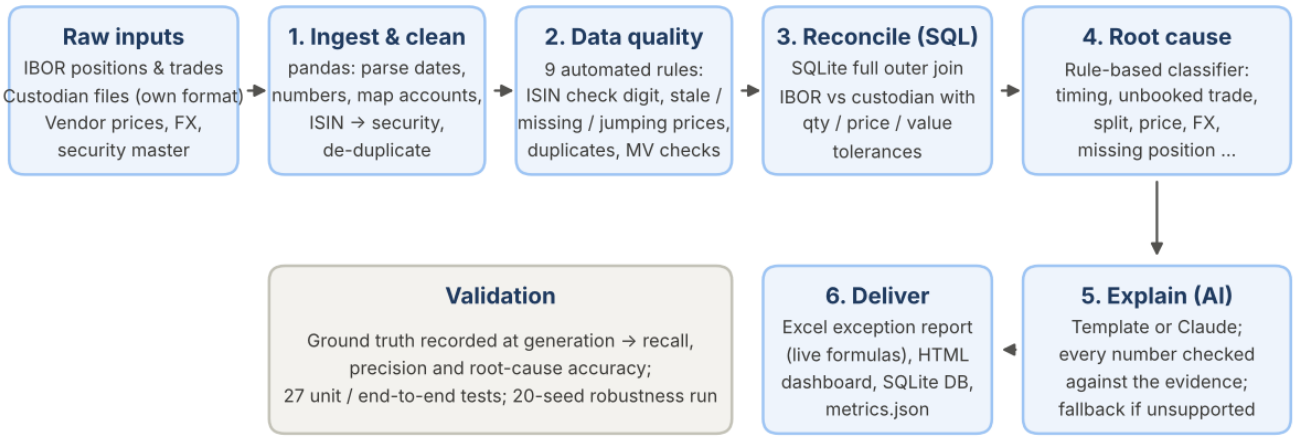
1. Business problem

An IBOR records positions on *trade date*; a custodian reports *settled* positions. On any day the two will disagree for legitimate reasons (trades not yet settled) and for real ones. An analyst has to separate the two, find the cause of each real break and route it to the right team (trade support, corporate actions, pricing, the custodian). The goals for this project were:

- **Data quality first:** catch bad inputs (invalid identifiers, duplicates, stale or erroneous prices) before they create noise.
- **Automated matching** with explicit, configurable tolerances for quantity, price and market value.
- **Root cause, not just a flag:** every break gets a category backed by evidence, a severity and an action.

- **Safe use of AI:** use an LLM to write explanations only if every figure it states can be traced to source data.
- **Measured, not claimed:** score the pipeline against known answers and test it.

2. Architecture



Each stage is a separate module in `recon/`; the reconciliation itself is plain SQL in `sql/reconcile_positions.sql`, and all thresholds live in `recon/config.py`.

3. Data and injected scenarios

Because custodian and IBOR data are confidential, a generator builds a realistic month: a 60-security master file (Canadian, US and German equities, Canadian bonds priced per 100 face value, ETFs) with ISO 6166-valid ISINs; prices following geometric Brownian motion; daily USD/CAD and EUR/CAD rates; three portfolios trading 2-5 times a day with T+1 settlement. Custodian files use a different layout from the internal model (DD/MM/YYYY dates, thousands separators, custodian account numbers, upper-case descriptions). The following problems are injected and written to a ground-truth file:

Scenario injected	What happens in the data	Expected outcome
Unprocessed 2-for-1 split	Custodian doubles the quantity on the ex-date; IBOR updates two days late	Corporate action not processed; price-jump DQ flag
Trade not booked	A settled buy is missing at the custodian for three days	Trade not booked at custodian
Stale vendor price	Vendor repeats one price for five days	Price difference; stale-price DQ flag
Missing vendor price	No price received; IBOR carries the last price forward	Price difference; missing-price DQ flag
Bad print (x100)	Vendor publishes an ETF price 100 times too high	Price difference; price-jump DQ flag
Wrong FX snapshot	Custodian converts one portfolio's USD holdings at a different rate	FX rate difference
Transfer-in not in IBOR	Securities received free of payment, booked in IBOR three days later	Position missing in IBOR
Row omitted / keying error	One position missing from a custodian file; one quantity off by 500	Position missing at custodian; unexplained quantity
Duplicate row / mistyped ISIN	Custodian file repeats a row; one ISIN has a wrong check digit	DQ flags; row de-duplicated and ISIN remapped by description

4. Data-quality rules

Rule	Logic
INVALID_ISIN	Format check plus ISO 6166 check digit (letters expanded to digits, Luhn algorithm)
UNKNOWN_SECURITY	Valid ISIN not present in the security master
DUPLICATE_RECORD	Same date, account and ISIN appears more than once in a custodian file
MISSING_PRICE	Held security with no vendor price that day
STALE_PRICE	Identical vendor price for 3+ consecutive business days
PRICE_JUMP	Price more than 20% away from the median of the prior 5 prices (median, so one bad print does not cause a second alert)
MV_INCONSISTENT	Quantity x price x multiplier differs from the reported market value
NEGATIVE_QUANTITY / MISSING_FX	Short position in a long-only portfolio; currency with no FX rate

5. Reconciliation and root-cause logic

Positions are joined on date, portfolio and security with a full outer join. A line breaks if the quantity differs by more than 0.0001, the price by more than 25 bps, or the CAD market value by more than the larger of CAD 5 and 1 bp. The same rule exists in Python, and a test asserts both give identical results on every line. Breaks are then classified in a fixed order, each step requiring positive evidence:

Order	Category	Evidence required
1	Timing - unsettled trade	Quantity gap equals the net of trades with trade date ≤ today < settlement date
2	Trade not booked at custodian	Gap equals one trade that settled in the last 14 days
3	Corporate action not processed	Custodian/IBOR quantity ratio is a split ratio (2, 3, 1/2 ...) and prices agree
4	Position missing in IBOR / at custodian	One side absent and nothing above explains it
5	Unexplained quantity	Any other quantity gap (escalated)
6	Price difference vs FX rate difference	Quantities agree; local value differs (price) or only CAD value differs (FX)

A line can carry a second problem behind the first (for example an unsettled trade *and* a bad price). Any price or FX gap behind another category is recorded as a secondary issue and stops the break from being auto-cleared. Severity is set from absolute CAD impact (Critical ≥ 1m, High ≥ 100k, Medium ≥ 10k), and the age of each break is tracked in business days.

6. AI explanations with a verification gate

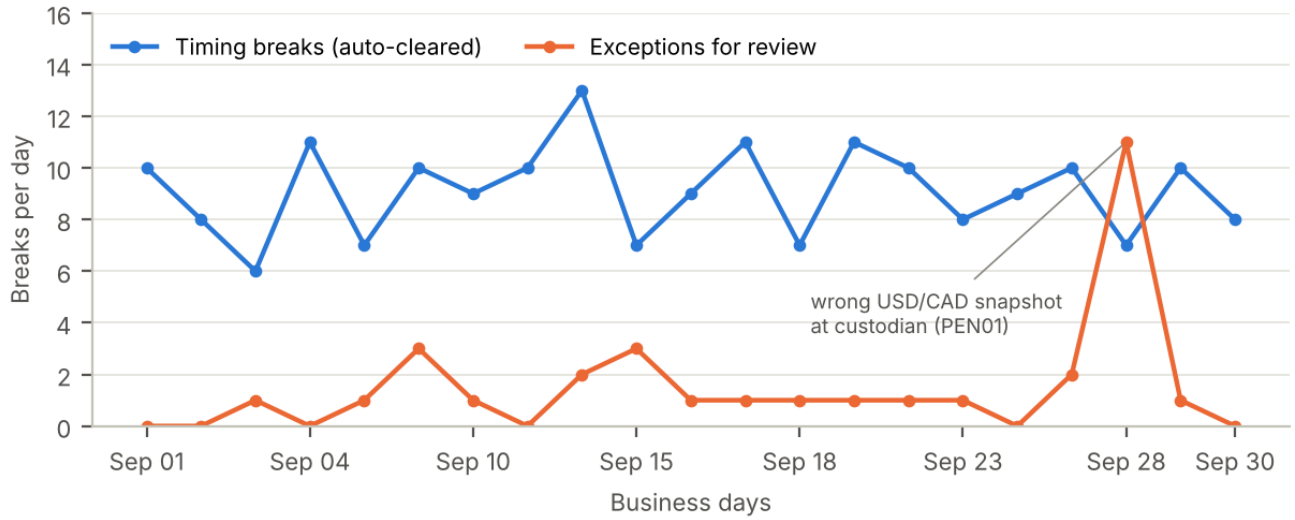
Every break carries an evidence record (quantities, prices, FX, pending or matched trades, DQ flags, impact). By default a deterministic template turns it into an explanation. With `--llm` the evidence is sent to Claude through the Messages API with an instruction to use only those facts. The reply is then checked: every date must appear in the evidence and every number must equal a source figure after rounding to the decimals shown. If any figure cannot be traced, the LLM text is rejected and the template is used, and the rejected tokens are logged. API errors also fall back to the template, so the pipeline never depends on the model being available or correct.

Example (generated): The custodian holds 12,200 units of CEHE (Cedar Health Inc.) in INS01 versus 6,100 in IBOR, a ratio of exactly 2 while both sides use the same price (80.0400). This pattern indicates a 2-for-1 split that IBOR has not processed; IBOR market value is understated by CAD 488,244.00. A price-jump data-quality flag was raised for the same date. *Action:* Ask Corporate Actions to process the 2-for-1 split in IBOR and confirm the ex-date.

Tests cover the gate directly: an answer quoting a made-up CAD amount or date is rejected, a correct answer is accepted, and a timeout falls back cleanly. The live API path needs the user's own key and was exercised through these tests with stubbed responses.

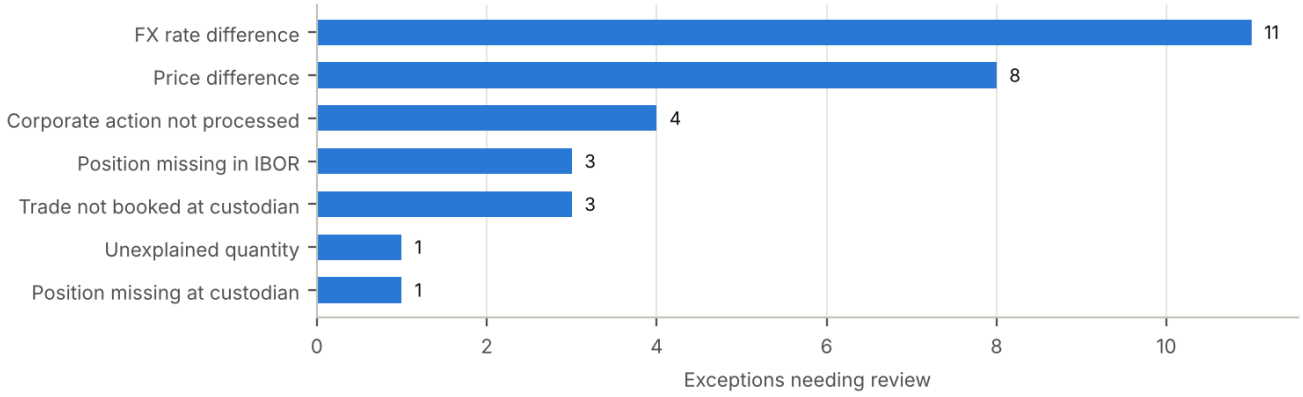
7. Results

Daily breaks



Timing breaks track trading volume and clear the next day. Exceptions stay low except on the day the custodian used a different USD/CAD rate, which affected every USD holding in one portfolio at once.

Exceptions by root cause



Accuracy against ground truth

Category	Expected	Correct
Timing - unsettled trade	191	191
Trade not booked at custodian	3	3
Corporate action not processed	4	4
Price difference	8	8
FX rate difference	11	11
Position missing in IBOR	3	3
Position missing at custodian	1	1
Unexplained quantity	1	1

Robustness check

To make sure the result is not a lucky draw, the whole generate-and-reconcile cycle was repeated with 20 random seeds (48,696 position lines, 4,544 breaks). Breaks missed: 0; false breaks: 0; root-cause accuracy 100%. The data-quality rules missed none of 200 injected issues but raised 2 false alerts (precision 99.0%): 2 stale-price flags on low-volatility ETFs whose two-decimal price genuinely did not change for three days. This is a real limitation of a fixed run-length rule and is listed under next steps.

8. Testing

- **Unit tests:** ISIN check digit on public ISINs, tolerance boundaries, each data-quality rule, each classifier branch including secondary issues and break ageing, and the AI verification gate.
- **End-to-end tests:** fresh data in a temporary folder; every injected break and DQ issue found; SQL and Python break rules agree on every line; duplicate removed and ISIN remapped; database queryable; generation reproducible from the seed.
- **Excel output:** 139 summary and trend formulas recalculated with zero errors and checked against the Python figures.

9. Limitations and next steps

- Synthetic data: scenarios are clean and one cause per break by construction. Real breaks often overlap; the secondary-issue flag is a first step.
- Stale-price rule: make the run length depend on volatility and price precision to remove the ETF false alerts.
- Scope: positions only. Extend to cash and transaction-level reconciliation, accrued interest for bonds, and private-market holdings.
- Operations: schedule the daily run, push exceptions to a ticket queue, and publish the SQLite tables to Power BI; track break ageing and resolution time over months.